ROYAL UNIVERSITY OF PHNOM PENH

Project Practicum I (PP)

FOUNDATIONS • ARCHITECTURE • UNIVERSITY APPLICATION

Retrieval-Augmented Generation (RAG)

Grounding Large Language Models in dynamic, verifiable institutional knowledge.

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PRESENTATION ROADMAP

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CORE PARADIGM

What is Retrieval-Augmented Generation?

The "Open-Book" Principle

Standard LLMs answer from memory. RAG consults external verified documents before generating output.

Hallucination Mitigation

Forces responses to cite retrieved passages, eliminating fabricated knowledge and outdated facts.

Dynamic Updatability

Knowledge bases refresh instantaneously by updating index files without costly model retraining.

THE 4-STAGE RUNTIME SEQUENCE

1. **Indexing**
   Raw university documents parsed and stored as vector embeddings
2. **Retrieval**
   Semantic similarity search fetches top-k matching contextual chunks
3. **Augmentation**
   User prompt is merged with fetched excerpts into an enriched prompt
4. **Generation**
   LLM generates grounded answers strictly backed by provided data

Origins & NeurIPS 2020 Paper

RESEARCH CONSORTIUM

Meta AI, UCL & NYU

Co-authored by Patrick Lewis et al. at Facebook AI Research in collaborative partnership with University College London and New York University.

SEMINAL PAPER

Knowledge-Intensive NLP

"Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks", combining pre-trained parametric models with non-parametric index memory.

PUBLICATION VENUE

NeurIPS 2020

Accepted at the 34th Conference on Neural Information Processing Systems, establishing the gold benchmark for enterprise retrieval agents.

TECHNICAL DIFFERENTIATION

Standard Generative AI vs. RAG Systems

Architectural Dimension	Standalone Generative AI	Retrieval-Augmented Generation
Knowledge Source	Internal training weights only	**Custom curated external documents**
Private University Data	No access to private campus databases	**Fully integrates private handbooks & records**
Knowledge Freshness	Fixed by pre-training cutoff date	**Instant updates when adding new files**
Accuracy & Grounding	Prone to convincing hallucinations	**Strictly answers based on retrieved context**

PRACTICAL IMPACT

What Is RAG Mainly Used For?

Private Data QA

Querying confidential enterprise knowledge bases, PDF libraries, legal archives, and internal institutional wikis with complete data privacy.

Helpdesks & Support

Powering intelligent customer service agents that answer policy, service, and administrative questions instantly with citations.

Academic Assistants

Supporting university students by indexing slide decks, textbooks, and syllabus rules into interactive 24/7 AI tutors.

SYSTEM CAPABILITIES

Three Core Strengths of RAG

01

Multi-Modal Data Intake

Ingest PDFs, university websites, raw lecture notes, slide presentations, and video transcripts into a single unified index.

• Formats: PDF, HTML, MD, DOCX

02

Configurable Retrieval

Full engineering control over chunk sizes, overlap tokens, top-k ranking, and vector similarity algorithms (Cosine / HNSW).

• Tunable chunking & search depth

03

Grounded & Verifiable

Answers are restricted to verified source text. When answers are missing, the system transparently says "I do not know."

• Zero fabrication • Traceable

APPLIED PROJECT PRACTICUM

Our Goal: The RUPP Student Chatbot



THE PROBLEM

RUPP students currently search disconnected PDFs, Facebook notice boards, and separate faculty sites to locate major lists and schedules.

THE SOLUTION

A centralized RAG agent updated each semester by simply uploading new handbook PDFs, answering with 100% verified campus data.

TARGET STUDENT QUERIES

- "How many majors are currently offered under Faculty of Engineering?"
- "What time do Data Science and Engineering classes meet on Fridays?"
- "Which faculty administers the Department of ITE?"

IMPLEMENTATION BLUEPRINT

Engineering Implementation Pipeline

01

Data Intake

Collect official RUPP PDF catalogs, timetable sheets, and university degree requirements.

02

Preparation

Chunk documents, compute dense vector embeddings, and store inside an indexed vector database.

03

Retrieval

Encode query into vectors, run cosine similarity search, and grab top-k relevant fragments.

04

LLM Synthesis

Prompt Gemini with retrieved context chunks to output grounded, accurate student answers.

Planned Stack • LangChain Orchestrator • Qdrant Vector Store • Google Gemini LLM

`Data verified with RUPP Admin`

Grounding Eliminates Hallucination

RAG converts LLMs from speculative predictors into deterministic retrieval query engines.

Cost-Efficient Continuous Updates

No re-training or parameter fine-tuning required to add next semester's timetables and courses.

Tangible Campus Value

Provides 10,000+ RUPP students instant, validated answers directly from official university policies.

Thank You!

Questions & Discussion

Department of Information Technology & Engineering
Faculty of Engineering, RUPP